

Una aplicación entre espacios vectoriales es lineal si y solo si su gráfica es un subespacio vectorial

Lema 1. Sean X, Y dos espacios vectoriales sobre K , entonces

$$f: X \longrightarrow Y \text{ es lineal} \iff \mathcal{G}(f) \text{ es un subespacio vectorial de } X \times Y.$$

Demostración: Veamos las dos implicaciones:

$\Rightarrow$ Basta ver que $\mathcal{G}(f)$ es cerrado bajo combinaciones lineales: $\forall x, y \in X, \lambda, \mu \in K :$

$$\lambda(x, f(x)) + \mu(y, f(y)) = (\lambda x + \mu y, \lambda f(x) + \mu f(y)) = (\lambda x + \mu y, f(\lambda x + \mu y)) \in \mathcal{G}(f).$$

$\Leftarrow$ Sean $(x, f(x)), (y, f(y)) \in \mathcal{G}(f)$ y $\lambda, \mu \in K$. Entonces,

$$\lambda(x, f(x)) + \mu(y, f(y)) = (\lambda x + \mu y, \lambda f(x) + \mu f(y)) \in \mathcal{G}(f).$$

Luego $\lambda f(x) + \mu f(y) = f(\lambda x + \mu y)$, es decir, f es lineal.

■

Referenciado en

- teo-grafica-cerrada

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.